

Feedback Control Systems		
Date	Week	Topic
13-Feb-2024	1	Feedback control systems, block diagrams, Linear system, Time-invariant system, Block diagram and reduction, Disturbance, Open loop, Closed loop
20-Feb-2024	2	Modeling of electrical and mechanical systems in the frequency domain, Electrical circuits, Mechanical systems: Translational, rotational
27-Feb-2024	3	Modeling of electromechanical systems in the frequency domain, Direct current (DC) motor, Modeling of dynamic models in the time domain (state space) / Pre-test
5-Mar-2024	4	Nonlinear system, EX: Pendulum Linearization
12-Mar-2024	5	Analysis of control systems in the time domain: 1st and 2nd order dynamic models
19-Mar-2024	6	Analysis of control systems in the time domain: Higher order and complex dynamic models, steady state errors
26-Mar-2024	7	root locus technique and properties / Midterm-1
2-Apr-2024	8	Controller design using root locus technique
9-Apr-2024		
16-Apr-2024	9	Analysis of control systems in the frequency domain: Bode diagram
23-Apr-2024	10	Analysis of control systems in the frequency domain: Nyquist diagram and stability criterion
30-Apr-2024	11	Analysis of control systems in the frequency domain: Relative stability Design light
7-May-2024	12	Analysis of control systems in the frequency domain: Its relationship with the time domain, Midterm-2
14-May-2024	13	Analysis and design of state space control systems
21-May-2024	14	Introduction to the design of digital control systems

Evaluation:

Pre Test	1%
Midterm-1	24%
Midterm-2	25%
Assignment (x2)	10%
Final Post test	10%
Final Exam	30%
Total	100%

VF Condition 15.00 $(Arasınav-1+Arasınav-2)/2 \geq 15$